Runs showing output:

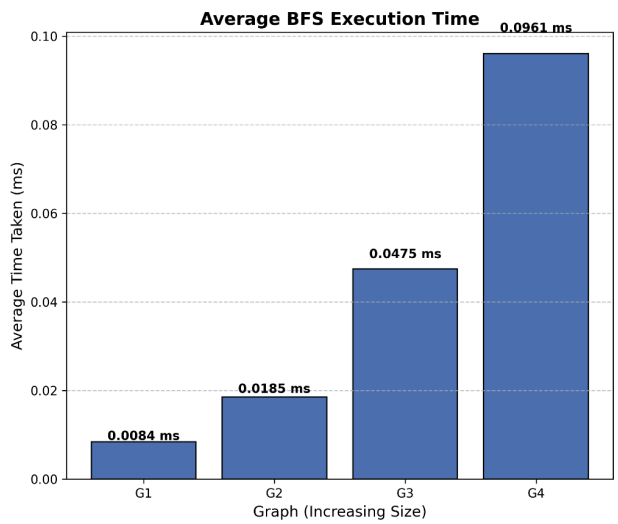
Graph	Source	Destination	Min-Hop Path	Time Taken (ms)
G1	28	13	28-19-7-13	0.0133
G1	14	9	14-20-9	0.0105
G1	9	1	9-7-2001	0.0072
G1	6	8	6-16-7-2-8	0.0091
G1	9	7	9-7	0.0049
G1	8	18	8-10-18	0.0085
G1	10	14	10-25-14	0.0082
G1	17	3	17-19-21-14-3	0.0099
G1	3	17	3-16-11-25-17	0.0102
G1	25	2	25-20-7-2	0.0098
G1	22	21	22-5-19-21	0.0076
G1	14	10	14-20-8-10	0.0101
G1	13	2	13-18-30-7-2	0.0086
G1	18	26	18-30-20-13-21-26	0.0105
G1	29	25	29-11-25	0.0072
G1	28	23	28-19-7-23	0.0067
G1	2	23	2-19-7-23	0.0096
G1	6	10	6-16-7-2-8-10	0.01
G1	11	12	11-25-12	0.0069
G1	5	21	5-19-21	0.0067
G1	7	24	7-2-24	0.0061
G1	17	5	17-11-25-20-5	0.0098
G1	30	24	30-20-24	0.0066
G1	9	24	9-24	0.0049
G1	13	20	13-18-30-20	0.0062
G1	AVERAGE			0.0084
G2	17	56	17-54-56	0.0148
G2	44	18	44-21-5-18	0.0204
G2	32	57	32-33-57	0.0186
G2	24	36	24-31-36	0.0128

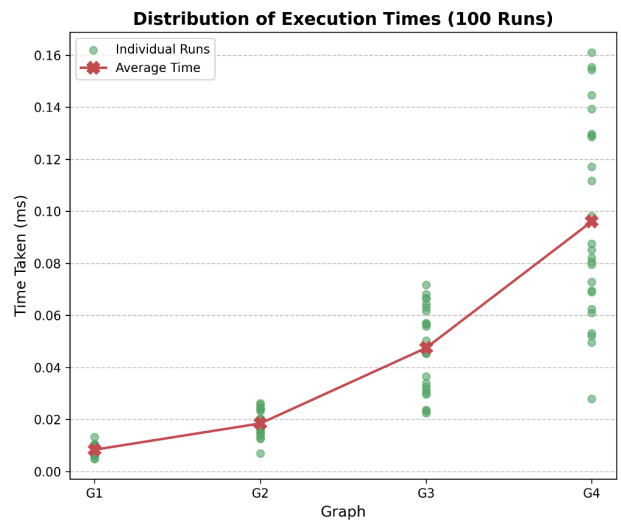
G2	51	59	51-4-59	0.0126
G2	18	5	18-48-5	0.0181
G2	21	3	21-5-11-3	0.0243
G2	17	25	17-51-25	0.0183
G2	6	59	6-27-59	0.014
G2	2	4	2-4	0.007
G2	43	19	43-50-19	0.0187
G2	50	39	50-28-39	0.0167
G2	27	52	27-51-49-52	0.0262
G2	44	48	44-21-5-48	0.0151
G2	55	1	55-34-1	0.0187
G2	33	13	33-19-13	0.0185
G2	20	38	20-11-49-38	0.0245
G2	43	55	43-3-55	0.0173
G2	59	8	59-51-38-8	0.0238
G2	3	28	3-42-28	0.0202
G2	57	16	57-18-26-16	0.0258
G2	31	21	31-23-21	0.017
G2	21	60	21-5-2-60	0.0232
G2	29	24	29-52-57-24	0.0197
G2	17	1	17-49-1	0.0158
G2	AVERAGE			0.0185
G3	9	34	9-38-34	0.0503
G3	82	30	82-30	0.0225
G3	12	76	12-45-76	0.0666
G3	97	20	97-106-20	0.0682
G3	9	120	9-120	0.0237
G3	22	75	22-92-75	0.0297
G3	90	107	90-97-107	0.0455
G3	77	46	77-84-81-46	0.0644
G3	104	83	104-56-83	0.0571
G3	55	94	55-13-94	0.0314
G3	106	85	106-45-85	0.0566
G3	3	88	3-1-88	0.0479

G3	41	99	41-18-99	0.046
G3	106	18	106-20-18	0.0567
G3	35	65	35-88-65	0.0365
G3	19	99	19-5-99	0.0325
G3	81	5	81-15-5	0.0453
G3	11	73	11-104-73	0.0618
G3	91	17	91-19-17	0.0302
G3	72	104	72-79-104	0.0666
G3	24	12	24-75-63-12	0.0631
G3	5	28	5-99-70-28	0.0717
G3	32	74	32-109-74	0.0339
G3	102	43	102-65-43	0.0558
G3	1	77	1-77	0.0233
G3	AVERAGE			0.0475
G4	124	182	124-240-182	0.0851
G4	193	45	193-61-45	0.0532
G4	190	203	190-209-203	0.0822
G4	202	160	202-128-160	0.0521
G4	81	183	81-75-56-183	0.1286
G4	209	81	209-25-128-81	0.1117
G4	134	59	134-67-188-59	0.1446
G4	74	124	74-143-124	0.0496
G4	170	179	170-175-156-179	0.1554
G4	52	9	52-49-44-9	0.1171
G4	97	13	97-13	0.0279
G4	12	40	12-182-224-40	0.0795
G4	132	184	132-188-184	0.0729
G4	96	48	96-176-48	0.0806
G4	148	232	148-97-13-232	0.161
G4	240	133	240-45-133	0.069
G4	240	116	240-219-77-116	0.1293
G4	46	23	46-201-23	0.0609

G4	54	48	54-187-48	0.0875
G4	163	156	163-202-156	0.0982
G4	13	209	13-166-37-209	0.1544
G4	15	128	15-25-128	0.0624
G4	42	1	42-210-1	0.0696
G4	100	148	100-177-72-148	0.1297
G4	26	157	26-39-14-157	0.1393
G4	AVERAGE			0.0961

Visualization:





Theoretical time analysis:

Let's say our graph G has nodes V and edges E. During the initialization phase, we iterate over each node in V and set its color, distance, and parent. This will naturally take O(V) time. The BFS will then push and pop each node it the queue exactly once. After each pop, the BFS iterates over every node in the popped node's adjacency list, which ultimately will amount to all the edges in the graph (E) and will take O(E) time. Therefore, the final time complexity will be O(E+V). This time complexity is for the worst-case scenario, where the destination does not exist or it's the last visited node. In my implementation, I made sure to stop the algorithm early when the destination is popped from the queue.

Results analysis:

Based on the experimental results, the runtime seems to be scaling linearly based on the input graph size. A good example can be seen when comparing G3 and G4. We know that:

G4.E + G4.V = 3440 and G3.V + G3.E = 1720.

G4 is exactly twice as big as G3, and its BFS runtime is also approximately twice as long.

The same pattern can be seen when transitioning from G1 to G2 or G2 to G3.

In conclusion, the experimental results are aligned with our theoretical analysis from earlier.